Project Health RAG Methodology

Phase 1 — how I'm scoring project health for the reporting agent

Before building the agent itself, I wanted to nail down exactly how a project's color gets decided, otherwise the system turns into a black box nobody can trust. The approach below splits project health into six signals, scores each on its own, and combines them into a single composite number. If a PM or a VP ever asks why a project is Amber and not Green, the answer should trace back to specific numbers, not a gut feeling.

The Six Signals

Signal	Weigh t	Green	Amber	Red
Schedule slippage	25%	<5% of tasks overdue	5–20% of tasks overdue	>20% of tasks overdue
Budget burn	20%	Actual spend within 10% of plan	10–25% over plan	>25% over plan
Milestone health	18%	All milestones on track	1 milestone delayed	2+ milestones delayed/missed
Blockers	12%	No open blockers	1–2 open, each <3 days old	3+ open, or any open >5 days
Critical path health	15%	No critical-path tasks overdue	1–2 critical-path tasks overdue	3+ critical-path tasks overdue
Stakeholder sentiment	10%	Notes indicate confidence, no concerns	Mixed or cautionary language	Explicit escalation or repeated dissatisfaction

Turning This Into One Number

Each signal becomes a number, 100 for Green, 60 for Amber, 20 for Red, multiplied by its weight and summed into one composite score. Above 80 is Green, 50–79 is Amber, below 50 is Red. It's deliberately simple: anyone can redo the math by hand and land on the same answer. An override sits on top of it: one Red signal caps the project at Amber; two or more Red signals force Red, regardless of what the composite number says, a straight average alone can hide a real problem (e.g. three blockers open two weeks) behind a couple of strong scores.

A Confidence Tag, Kept Separate From the Color

Every report also carries a confidence tag: High, Medium, or Low based on how much of the source data was actually filled in (roughly above 90%, 70–90%, and below 70% of expected fields). A project can come back Green, but if half its expected fields were blank, that Green deserves less trust kept as its own flag rather than letting spreadsheet gaps quietly distort the color itself.

Assumptions I'm Making About the Data

1.A signal with no data source at all for a project (no Comments field, no budget/cost field, etc.) is dropped and its weight redistributed proportionally across the remaining computable signals, rather than guessing or forcing a placeholder score.
2.A task counts as overdue if its Start/End window has passed and it isn't marked 100% complete; Blockers only count if not already resolved/closed.
3.Critical-path health uses each task's Total Float (schedule slack, in days) directly, treating effectively-zero float as critical-path, found by auditing every real column and discovering Total Float is 99.8–100% populated yet was completely unused, even though an overdue task with 100 days of float barely threatens the project while one with zero float directly delays the end date.
4.If fewer than 4 of the 6 signals are computable at all, the project is marked Insufficient Data rather than forced into a color.
5.The per-task drill-down (which individual tasks are driving a project's status) deliberately sticks to the same reliably-populated fields : Status, Start/End Date, % Complete, At Risk flag, Total Float, rather than Baseline Start/Finish or Variance, which were tried and then dropped: the reasoning was harder to defend simply, and those two fields were never as consistently populated as the others.
6.These weights and thresholds are a starting point, worth recalibrating once there are a few real reporting cycles to check them against.